

NOTES FOR MATH 3150, WEEK 4

The rules discussed so far simplify our calculations considerably, reducing them to a few basic limits (that we try to remember).

1. BASIC LIMITS

1. Ratio of polynomials. Suppose that $P(x) = a_0x^k + a_1x^{k-1} + \dots + a_k$ and $Q = b_0x^k + b_1x^{k-1} + \dots + b_k$ are polynomials of degree k , then

$$\lim_{n \rightarrow \infty} \frac{P(n)}{Q(n)} = \frac{a_0}{b_0}.$$

Proof. We have

$$\begin{aligned} \frac{P(n)}{Q(n)} &= \frac{a_0n^k + a_1n^{k-1} + \dots + a_k}{b_0n^k + b_1n^{k-1} + \dots + b_k} \\ &= \frac{a_0 + a_1\frac{1}{n} + \dots + a_k\frac{1}{n^k}}{b_0 + b_1\frac{1}{n} + \dots + b_k\frac{1}{n^k}}, \end{aligned}$$

where in the second line we divided both sides by n^k . Since $\frac{1}{n} \rightarrow 0$, by properties of limits we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(a_0 + a_1\frac{1}{n} + \dots + a_k\frac{1}{n^k} \right) &= a_0 \\ \lim_{n \rightarrow \infty} \left(b_0 + b_1\frac{1}{n} + \dots + b_k\frac{1}{n^k} \right) &= b_0. \end{aligned}$$

By properties of limits we conclude that

$$\lim_{n \rightarrow \infty} \frac{P(n)}{Q(n)} = \frac{a_0}{b_0}.$$

□

2. Exponentials. For any $|a| < 1$ we have

$$\lim_{n \rightarrow \infty} a^n = 0.$$

Proof. Instead of showing that $|a|^n$ is small, we show that $\left(\frac{1}{|a|}\right)^n$ is large.

In calculus we have seen often that exponentials grow faster than polynomials.

The following inequality is a simple example of this fact: $(1+x)^n \geq 1+nx$, for all $x \geq 0$.

This follows immediately from the binomial theorem, because

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k \geq \binom{n}{0} + \binom{n}{1} x = 1 + nx.$$

We apply the above observation to our context. Since $\frac{1}{|a|} > 1$, let

$$x = \frac{1}{|a|} - 1 > 0,$$

so that

$$\frac{1}{|a|} = 1 + x.$$

By the above inequality we have

$$\begin{aligned} \frac{1}{|a|^n} &= (1+x)^n \\ &\geq 1 + nx \\ &= 1 + n \left(\frac{1}{|a|} - 1 \right). \end{aligned}$$

Therefore, we have proved that

$$|a|^n \leq \frac{1}{1 + n \frac{1-|a|}{|a|}}.$$

Since the right side converges to zero, this proves the result. $\square$

3. Geometric progression. For any $|a| < 1$ we have

$$\lim_{n \rightarrow \infty} (1 + a + a^2 + \dots + a^n) = \frac{1}{1-a}.$$

Proof. Recall the formula

$$1 + a + a^2 + \dots + a^n = \frac{1 - a^{n+1}}{1 - a}.$$

Now take limit in $n \rightarrow \infty$ and use the previous result. $\square$

4. A limit. We have

$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0.$$

Proof. We cannot use limit laws, because the limit $\lim_{n \rightarrow \infty} \sin n$ does not exist. Note that

$$\left| \frac{\sin n}{n} \right| \leq \frac{1}{n}.$$

Now, either proceed by definition (and use that $\frac{1}{n} < \varepsilon$ for any $n > N = \frac{1}{\varepsilon}$), or apply the squeeze theorem. $\square$

2. THE INFINITY SYMBOLS

We use $-\infty$ and ∞ to understand unbounded quantities.

For example, if a set S is unbounded above, we say that $\sup S = \infty$. Similarly, if S is unbounded below, we say $\inf S = -\infty$.

Definition 1. (1) We say that a sequence $(x_n)_{n \geq 1}$ has

$$\lim_{n \rightarrow \infty} x_n = \infty$$

if for any $M > 0$ there exists N so that

$$x_n > M$$

for all $n > N$.

(2) We say that

$$\lim_{n \rightarrow \infty} x_n = -\infty$$

if

$$\lim_{n \rightarrow \infty} (-x_n) = \infty.$$

Examples:

1. Prove that

$$\lim_{n \rightarrow \infty} \frac{n^4 - 4n}{n^2} = \infty.$$

Proof. For any $M > 0$ large enough, we want to show there exists $N > 0$ so that

$$\frac{n^4 - 4n}{n^2} > M,$$

for all $n > N$. Instead of solving this in n directly, we note some preliminary estimates first. For example, for any $n \geq 2$ we have

$$n^4 - 4n > n^3,$$

because

$$\begin{aligned} n^4 - n^3 &= n^3(n - 1) \\ &\geq n^3 \\ &\geq 4n. \end{aligned}$$

Therefore,

$$\frac{n^4 - 4n}{n^2} \geq n,$$

which means that we can simply define $N = n$ and then

$$\frac{n^4 - 4n}{n^2} > M,$$

for all $n > N$. □

A fundamental property of a sequence is to be always increasing (or always decreasing). Such sequences are called monotone.

3. MONOTONE SEQUENCE

We start with the definition.

Definition 2. Let $(x_n)_{n \geq 1}$ be a sequence of real numbers.

- (1) $(x_n)_{n \geq 1}$ is called *increasing* if $x_n \leq x_{n+1}$ for all $n \geq 1$.
- (2) $(x_n)_{n \geq 1}$ is called *decreasing* if $x_n \geq x_{n+1}$ for all $n \geq 1$.
- (3) $(x_n)_{n \geq 1}$ is called *monotone* if it is either increasing or decreasing.

Examples:

- 1. $x_n = \frac{1}{n}$ is decreasing.
- 2. $x_n = \frac{n}{n+1}$ is increasing.

Indeed, $x_n \leq x_{n+1}$ is equivalent to $\frac{n}{n+1} \leq \frac{n+1}{n+2}$. In turn, this is equivalent to $n(n+2) \leq (n+1)^2$. After simplification, we see that this is true.

- 3. $x_n = \frac{(-1)^n}{n}$ is not monotonic.

Indeed, $x_{2n} = \frac{1}{2n}$, $x_{2n+1} = -\frac{1}{2n+1}$, and $x_{2n+2} = \frac{1}{2n+2}$. Therefore, $x_{2n} > x_{2n+1}$ and $x_{2n+2} > x_{2n+1}$, so the sequence is neither increasing nor decreasing.

We are ready to prove the following fundamental result.

Theorem 3 (Monotone Convergence Theorem). *Bounded monotone sequences are convergent.*

Proof. Without loss of generality, assume that $(x_n)_{n \geq 1}$ is increasing.

Let

$$S = \{x_n : n \geq 1\}.$$

The set S is bounded (because $(x_n)_n$ is bounded), so it has a supremum by the Completeness Axiom. Let

$$M = \sup S.$$

As M is an upper bound, we have

$$(1) \quad x_n \leq M$$

for all $n \geq 1$.

A property of supremum tells us that for any $\varepsilon > 0$ there exists $x_N \in S$ so that

$$M < x_N + \varepsilon.$$

Since $(x_n)_{n \geq 1}$ is increasing, this implies that

$$(2) \quad M < x_n + \varepsilon$$

for all $n \geq N$.

Then (1) and (2) prove that

$$|M - x_n| \leq \varepsilon$$

for all $n \geq N$. Thus the sequence $(x_n)_{n \geq 1}$ converges to M . $\square$

Examples:

1. **Iterated roots.** Let $(x_n)_{n \geq 1}$ be defined by $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + x_n}$. Prove that $(x_n)_{n \geq 1}$ converges to $L = 2$.

Proof. Clearly, $x_n \geq 0$ for all n .

We check if $(x_n)_{n \geq 1}$ is a monotone sequence. We will actually prove that the sequence is increasing.

Note that $x_{n+1} \geq x_n$ if and only if $\sqrt{2 + x_n} \geq x_n$. Squaring up, this is equivalent to $x_n^2 - x_n - 2 \leq 0$. Factoring out, this is equivalent to

$$(x_n - 2)(x_n + 1) \leq 0.$$

Since $x_n \geq 0$, we conclude that

$$x_{n+1} \geq x_n$$

if and only if

$$x_n \leq 2.$$

We prove by induction that

$$x_n \leq 2 \text{ for all } n \geq 1.$$

Clearly, $x_1 \leq 2$. Assume that $x_n \leq 2$ and prove that $x_{n+1} \leq 2$. Indeed,

$$\begin{aligned} x_{n+1} &= \sqrt{2 + x_n} \\ &\leq \sqrt{2 + 2} \\ &= 2. \end{aligned}$$

In conclusion, $x_n \leq 2$ for all $n \geq 1$. By the above discussion, we have that $(x_n)_{n \geq 1}$ is an increasing sequence.

Moreover, the sequence is bounded because

$$0 < x_n \leq 2$$

for all $n \geq 1$.

This proves that $(x_n)_{n \geq 1}$ is convergent, let L be its limit. Clearly,

$$\lim_{n \rightarrow \infty} x_{n+1} = L$$

as well.

Taking limit in

$$x_{n+1} = \sqrt{2 + x_n}$$

and using limit properties, we conclude that

$$L = \sqrt{2 + L}.$$

Solving this equation we get $L = 2$. □

Remark. An alternative way to proving x_n is increasing is to plug in the recurrence formula for both x_{n+1} and x_n and apply induction (this works for some, but not all, problems).

So, we see that $x_{n+1} \geq x_n$ is equivalent to $\sqrt{2 + x_n} \geq \sqrt{2 + x_{n-1}}$. Squaring up, this is equivalent to $x_n \geq x_{n-1}$.

This means that we can proceed by induction and prove that $x_{n+1} \geq x_n$ for all $n \geq 1$. Note that you still have to prove that x_n is bounded above in order to use the Monotone Convergence Theorem!

2. Decimals.

Suppose $K \geq 0$ is an integer and $d_i \in \{0, 1, 2, \dots, 9\}$. Define

$$x_n = K + \frac{d_1}{10} + \frac{d_2}{10^2} + \dots + \frac{d_n}{10^n}.$$

Then $(x_n)_n$ is a bounded increasing sequence.

It is bounded below by K and above by $K + 1$.

Hence $(x_n)_n$ converges to a limit that we traditionally denote

$$x = K.d_1d_2d_3\dots$$

For example,

$$\begin{aligned} 0.33333\dots &= \lim_{n \rightarrow \infty} \left(\frac{3}{10} + \frac{3}{10^2} + \dots + \frac{3}{10^n} \right) \\ &= \frac{3}{10} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{10} + \dots + \frac{1}{10^n} \right) \\ &= \frac{3}{10} \frac{1}{1 - \frac{1}{10}} \\ &= \frac{1}{3}. \end{aligned}$$

Another example:

$$\begin{aligned} 0.9999\dots &= \lim_{n \rightarrow \infty} \left(\frac{9}{10} + \frac{9}{10^2} + \dots + \frac{9}{10^n} \right) \\ &= \frac{9}{10} \frac{1}{1 - \frac{1}{10}} \\ &= 1. \end{aligned}$$